a. The excess glucose is excreted in the urine, reducing the likelihood of weight gain since they are fat free.
b. The excess glucose is converted to triacylglycerols in the liver, transported to adipose tissue, and stored as fat, leading to weight gain.
c. The excess glucose is converted into additional glycogen beyond the normal storage capacity, accumulating in the liver and muscles and leading to weight gain.
d. The excess glucose is converted into amino acids and stored in muscle tissue, increasing body mass and leading to weight gain.

The correct answer is: The excess glucose is converted to triacylglycerols in the liver, transported to adipose tissue, and stored as fat, leading to weight gain.

35. During a nutrition education session, a nurse explains to a patient that some vitamins are absorbed more efficiently when consumed with foods containing dietary fat. Which of the following best explains this advice?
    a. Fat-soluble vitamins such as A, D, E, and K require the presence of dietary fat for optimal absorption and bioavailability.
    b. Water-soluble vitamins such as B-complex and C depend on fat intake for absorption in the small intestine.
    c. All vitamins are absorbed equally regardless of the presence or absence of dietary fat.
    d. Fat-soluble vitamins such as A, D, E, K, B_3, and folate require dietary fat for efficient absorption and bioavailability.

The correct answer is: Fat-soluble vitamins such as A, D, E, and K require the presence of dietary fat for optimal absorption and bioavailability.

36. Fructose is a naturally occurring monosaccharide that differs structurally from glucose and is commonly encountered in the human diet. Which of the following foods would be expected to provide fructose as a major carbohydrate source?
    a. Brown rice, oats, and whole wheat bread
    b. Apples, honey, and table sugar
    c. Chicken breast, eggs, and fish
    d. Cheese, yogurt, and butter

The correct answer is: Apples, honey, and table sugar

37. Which pair of carbohydrates demonstrates the most significant structural difference in hydroxyl group orientation, where one is less stable than the other and therefore rapidly converted by the liver to achieve greater stability for use as the body's primary fuel source?
    a. Glucose and Fructose
    b. Galactose and Glucose
    c. Maltose and Lactose
    d. Amylose and Amylopectin

The correct answer is: Galactose and Glucose

38. According to Ghana's Food-Based Dietary Guidelines, what percentage by weight of foods consumed daily should come from fruits?
    a. 10%
    b. 12.5%
    c. 15.3%
    d. 20%

The correct answer is: 15.3%

39. What is the primary factor that distinguishes the glycemic response of highly processed foods from raw foods with comparable fiber content?
    a. The speed of digestion and absorption
    b. The total carbohydrate content
    c. The presence of artificial additives
    d. The moisture content

The correct answer is: The speed of digestion and absorption

40. A patient regularly consumes white rice (high glycemic index) as a staple food and experiences postprandial hyperglycemia. Which of the following dietary modifications would be MOST effective in reducing the glycemic response to this meal?
    a. Replace white rice with brown rice and increase the portion size to maintain satiety
    b. Reduce the portion size of white rice and combine it with legumes, vegetables, and a source of healthy fats such as avocado or nuts
    c. Consume the white rice on an empty stomach to maximize insulin sensitivity and follow with protein 2 hours later
    d. Maintain the current portion size but add a whey protein supplement immediately before the meal to stimulate insulin secretion

The correct answer is: Reduce the portion size of white rice and combine it with legumes, vegetables, and a source of healthy fats such as avocado or nuts

41. Which B vitamin deficiency causes pellagra? **Vitamin B-complex (NIACIN)**
42. Consider these 4 minerals: Calcium, Sodium, Iron, Sulphur. Which one (1) is needed in less than 100mg per day? **Iron**
43. How many fat-soluble vitamins are there? **4 (vitamin A,D,E,K)**
44. State one scanning method for accessing body composition…. **Dual-energy X-ray absorptiometry**.
45. An alcoholic consumes 100ml of alcohol early in the morning on an empty stomach. How much energy is available to this alcoholic (take 1ml = 0.789g)?

If 1ml = 0.789g
100ml = ?
100ml x 0.789g = 78.9g
So energy from alcohol is……..
Alcohol provides 7kcal/g
78.9g x 7kcal/g = 552.3kcal

46. Mention one of the three underlying causes of malnutrition based on the UNICEF conceptual framework .... **Disease / inadequate dietary intake**
47. Which quantitative dietary assessment method relies on the respondent's memory? **24 hour dietary recall**
48. A 4-year-old boy's BMI has been calculated, what is the ideal z-score/index for determining if the child is obese or not? **BMI-for-Age**
49. Mention one of the two main types of malnutrition there are? **Undernutrition or Overnutrition**
50. Mention one of the two essential fatty acids that the body must obtain from food? **Omega-3 or Omega-6**
51. Which anthropometric index can be used to classify a 1-year-old female as either overweight or obese? **Weight-for-Length**
52. How many macro-minerals are there? **7**
53. How many trace minerals are there? **9**
54. State one anatomical site used for skinfold thickness measurement **Biceps/Triceps/Midaxilliary/Subscapular skinfold**
55. The height of a 73kg young woman is 168cm, what is her BMI?
    Weight = 73kg
    Height = 168/100 = 1.68m (change to meters)
    BMI = 73kg/$(1.68)^2$ = 25.86kg/m
56. Mention one of the two immediate causes of malnutrition based on the UNICEF conceptual framework…**Inadequate dietary intake**
57. Which quantitative dietary assessment method relies on the respondent's memory? **24 hour dietary recall**
58. Among the 3 quantitative dietary methods, which one is referred to as the gold standard because it can be used to assess the actual nutrient intakes of individuals? **24 hour recall**
59. What are the two essential fatty acids that the body must obtain from food? **Omega-3 and Omega-6**
60. Mention one anthropometric measurement…… **Age or Length or Height or Weight**
61. Mention one anthropometric index………… **WFH or WFL or** WFA
62. The imaginary plane which passes through the external auditory meatus and over the top of the lower bone of the eye socket is called **Frankfurt Plane**
    NOTE; when the baby is lying on the infantometer the plane becomes vertical and when the baby is standing it becomes horizontal

63. A nurse at a pediatric clinic measures the recumbent length of a 20-month-old child and records 78m, what value for length should be plotted on the weight-for-length z-score ... **78cm**
64. What is the name of the device used to measure recumbent length .... **Infantometer**
65. The improved absorption of fat-soluble nutrients In the body due to the presence of dietary fats Is known as **Enhanced Bioavailability**
66. The three main types of lipids found in foods and the body are
    i. **Sterols**
    ii. **Phospholipids**
    iii. **Triglycerides**
67. A triacylglycerol is formed when one molecule of **GLYCEROL** is joined by three **FATTY ACIDS**